

Sync or Sink — Labor Market Analysis

Project Overview

Sync or Sink — Labor Market Analysis is an end-to-end data analytics project that explores the global labor market using salary and employment data. The project combines data preprocessing, cleaning, analysis, data modeling, and interactive visualization to uncover insights about salaries, education, experience, industries, and work modalities.

Using modern data analytics tools, the project transforms raw data into meaningful business intelligence that helps individuals, recruiters, and organizations better understand labor market trends and make informed decisions.

The project follows a complete analytics workflow:

Data Collection

Data Cleaning & Preprocessing

Data Modeling

Data Analysis

Interactive Dashboard Development

Business Insights & Reporting

Project Objectives

Analyze salary distribution across different industries.

Study the impact of education on compensation.

Measure the effect of experience and seniority on salary.

Compare Remote, Hybrid, and On-site work models.

Discover salary gaps between different job categories.

Build interactive dashboards that support business decision-making.

Dataset

The project is based on a dataset containing over 250,000 salary records from different industries and countries.

The dataset includes information such as:

Job Title

Industry

Department

Years of Experience

Education Level

Certifications

Salary

Bonus

Tax Rate

Work Modality

Growth Percentage

Project Workflow

Raw Dataset

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Data Cleaning (Python & Excel)

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Data Preprocessing

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Data Modeling (Star Schema)

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Power BI

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Interactive Dashboard

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Business Insights

★ Data Model — Star Schema

The project uses a Star Schema to improve query performance and simplify data analysis.

Fact Table

Fact_SalaryPerformance

Contains numerical business metrics including:

Annual Salary

Net Compensation

Tax Rate

Bonus

Salary Growth

Dimension Tables

Dim_Job

Job Title

Industry

Department

Work Modality

Required Skills

Dim_Experience

Years of Experience

Seniority Level

Previous Roles

Dim_Education

Degree Level

Certifications

Field of Study

Tools & Technologies

Tool Purpose

Python (Pandas, NumPy) Data cleaning & preprocessing

Microsoft Excel Data validation, formatting & initial cleaning

Power BI Desktop Data visualization & dashboard development

Power Query Data transformation inside Power BI

DAX Measures, KPIs & calculated columns

Star Schema Data warehouse modeling

Dashboard Features

Salary Analysis

Average Salary

Median Salary

Salary Distribution

Salary Growth

Job Insights

Highest Paying Jobs

Salary by Industry

Salary by Department

Top Skills

 Education Analysis

Salary by Degree

Salary by Certification

Field of Study Comparison

 Experience Analysis

Salary by Experience

Seniority Analysis

Career Growth

 Work Analysis

Remote vs Hybrid vs On-site

Work Modality Distribution

 Interactive Features

Dynamic Filters

Drill-through Pages

Cross-filtering

KPI Cards

Trend Analysis

 Key Insights

The dashboard helps answer questions such as:

Which industries pay the highest salaries?

Does education significantly affect income?

How much does experience increase salary?

Which work modality offers the best compensation?

What skills are most associated with high-paying jobs?

Getting Started

Prerequisites

Microsoft Power BI Desktop

Python 3.x (optional, for preprocessing)

Installation

git clone https://github.com/mahmoudtarek10/Sync_or_Sink.git

Open:

Sync_or_Sink.pbix

Explore:

Model View

Report View

Dashboard Pages

Repository Structure

Sync_or_Sink/

|

|— Dashboard/

|

|— Sync_or_Sink.pbix

|

|— Dataset/

```
| |— Raw Data
| |— Clean Data
|
|— Python/
| |— Data_Preprocessing.ipynb
|
|— Images/
|
|— README.md
|
|— LICENSE
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Team

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License

This project is developed for educational purposes.